In [1]:
```
import numpy as np
```

In [2]:
```
ones_arr = np.ones((5,5))
```

In [3]:
```
ones_arr
```

Out[3]:
```
array([[1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1.]])
```

In [4]:
```
ones_arr = np.ones((5,5), dtype=int)
```

In [5]:
```
ones_arr
```

Out[5]:
```
array([[1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1]])
```

In [6]:
```
zeros_arr = np.zeros((3,3), dtype=int)
```

In [7]:
```
zeros_arr
```

Out[7]:
```
array([[0, 0, 0],
       [0, 0, 0],
       [0, 0, 0]])
```

In [8]:
```
ones_arr
```

Out[8]:
```
array([[1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1],
       [1, 1, 1, 1, 1]])
```

In [9]:
```
ones_arr * 255
```

Out[9]:
```
array([[255, 255, 255, 255, 255],
       [255, 255, 255, 255, 255],
       [255, 255, 255, 255, 255],
       [255, 255, 255, 255, 255],
       [255, 255, 255, 255, 255]])
```

In [10]:
```
import matplotlib.pyplot as plt
```

In [11]:
```
%matplotlib inline
```

In [12]:
```
from PIL import Image  # python imaging library
```

In [13]:
```
elephant_img = Image.open(r'C:\Users\Admin\elephant.jpg')
```

In [14]:
```
elephant_img
```

Out[14]:

In [15]:
```
type(elephant_img)
```

Out[15]:
```
PIL.JpegImagePlugin.JpegImageFile
```

In [16]:
```
elephant_arr = np.asarray(elephant_img)
elephant_arr
```

Out[16]:
```
array([[[195, 207, 223],
        [191, 203, 219],
        [186, 198, 214],
        ...,
        [182, 196, 207],
        [181, 195, 206],
        [181, 195, 206]],

       [[193, 205, 221],
        [189, 201, 217],
        [185, 197, 213],
        ...,
        [181, 195, 206],
        [181, 195, 206],
        [180, 194, 205]],

       [[191, 203, 219],
        [188, 200, 216],
        [183, 195, 211],
        ...,
        [180, 194, 205],
        [180, 194, 205],
        [179, 193, 204]],

       ...,

       [[149,  99,  62],
        [141,  92,  52],
        [158, 106,  66],
        ...,
        [167, 124,  73],
        [160, 119,  67],
        [151, 111,  59]],

       [[164, 115,  75],
        [172, 123,  82],
        [162, 111,  68],
        ...,
        [188, 140,  91],
        [183, 137,  87],
        [182, 139,  88]],

       [[147,  98,  57],
        [179, 130,  87],
        [159, 108,  63],
        ...,
        [156, 104,  54],
        [162, 112,  61],
        [176, 129,  77]]], shape=(408, 612, 3), dtype=uint8)
```

In [17]:
```
type(elephant_arr)
```

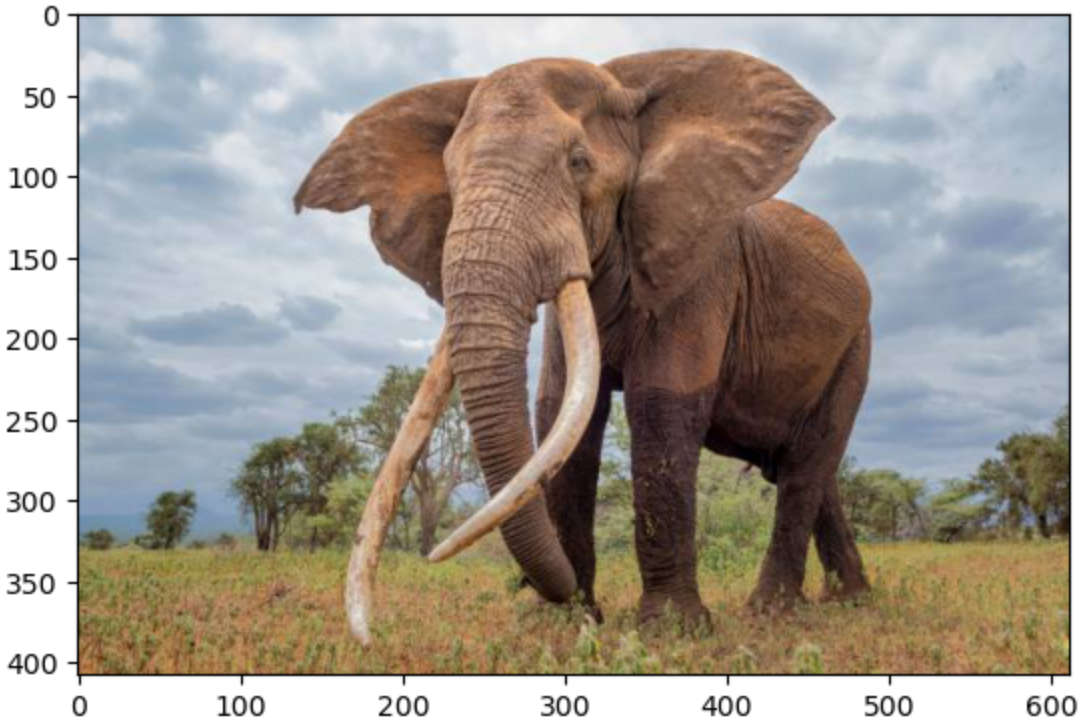
Out[17]:
```
numpy.ndarray
```

In [18]:
```
elephant_arr.shape
```

Out[18]: (408, 612, 3)

In [19]:
```
plt.imshow(elephant_arr)
```

Out[19]: <matplotlib.image.AxesImage at 0x2a9dd0ff380>



In [20]:
```
elephant_red = elephant_arr.copy()
```

In [21]:
```
elephant_red
```

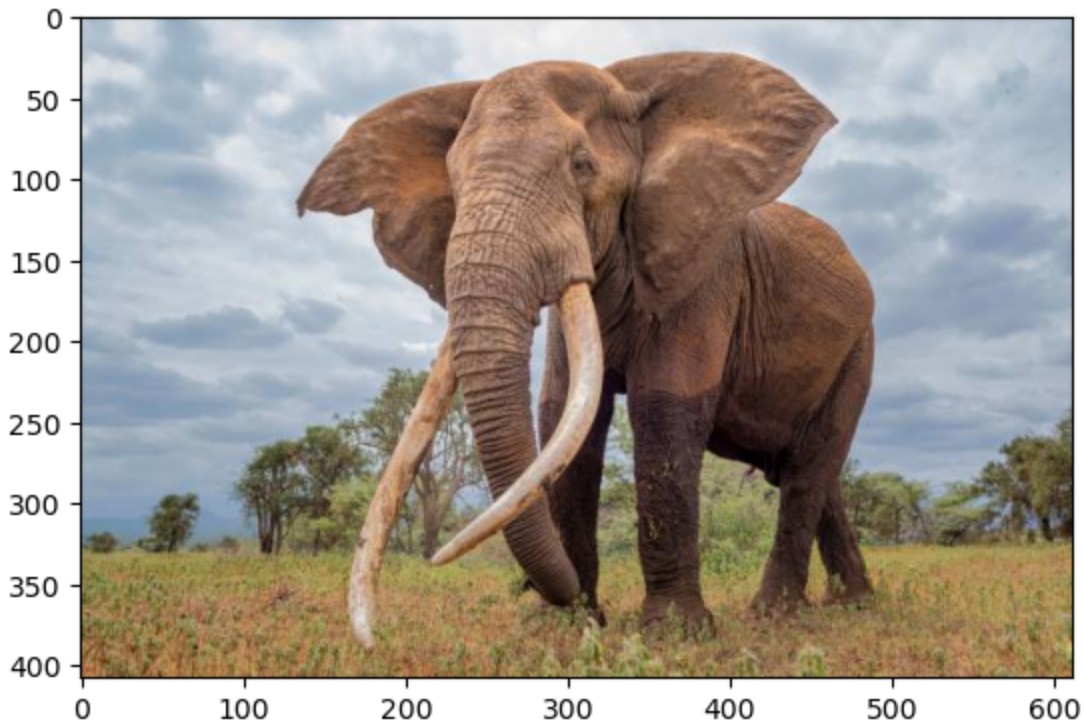
Out[21]:
```
array([[[195, 207, 223],
        [191, 203, 219],
        [186, 198, 214],
        ...,
        [182, 196, 207],
        [181, 195, 206],
        [181, 195, 206]],

       [[193, 205, 221],
        [189, 201, 217],
        [185, 197, 213],
        ...,
        [181, 195, 206],
        [181, 195, 206],
        [180, 194, 205]],

       [[191, 203, 219],
        [188, 200, 216],
        [183, 195, 211],
        ...,
        [180, 194, 205],
        [180, 194, 205],
        [179, 193, 204]],

       ...,

       [[149,  99,  62],
        [141,  92,  52],
        [158, 106,  66],
        ...,
        [167, 124,  73],
        [160, 119,  67],
        [151, 111,  59]],

       [[164, 115,  75],
        [172, 123,  82],
        [162, 111,  68],
        ...,
        [188, 140,  91],
        [183, 137,  87],
        [182, 139,  88]],

       [[147,  98,  57],
        [179, 130,  87],
        [159, 108,  63],
        ...,
        [156, 104,  54],
        [162, 112,  61],
        [176, 129,  77]]], shape=(408, 612, 3), dtype=uint8)
```

In [22]:
```
elephant_arr == elephant_red
```

Out[22]:
```
array([[[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       ...,

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]],

       [[ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True],
        ...,
        [ True,  True,  True],
        [ True,  True,  True],
        [ True,  True,  True]]], shape=(408, 612, 3))
```

In [23]:
```
plt.imshow(elephant_red)
```

Out[23]:
```
<matplotlib.image.AxesImage at 0x2a9de1c9310>
```

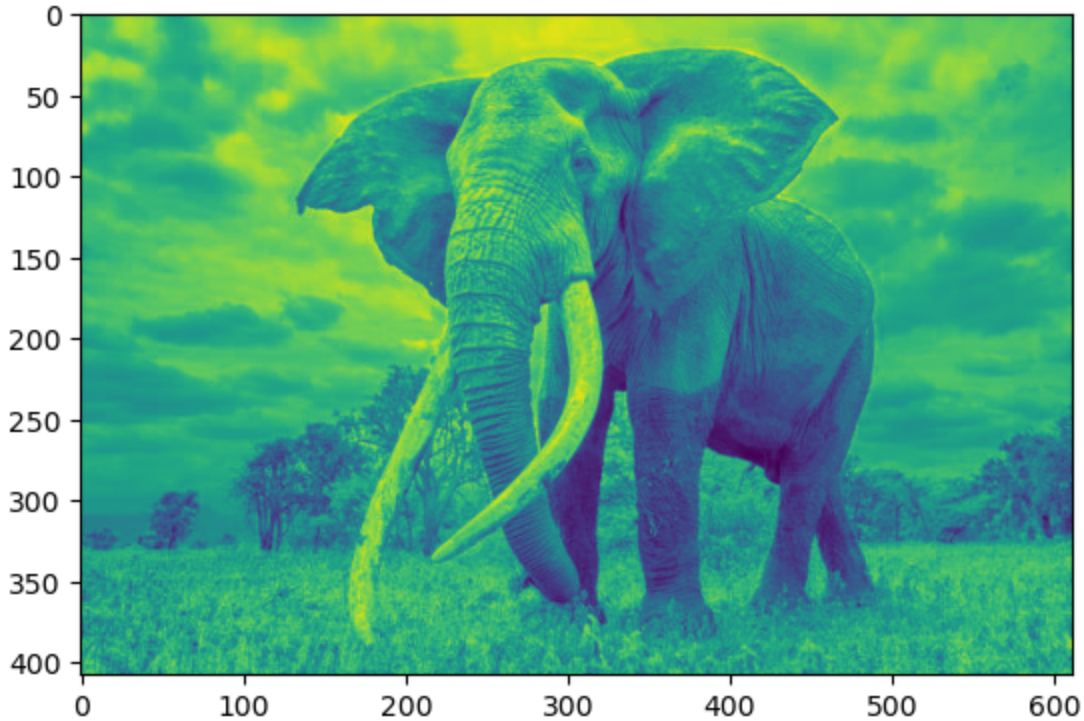


In [24]:
```
elephant_red.shape
```

Out[24]:
```
(408, 612, 3)
```

In [25]:
```
plt.imshow(elephant_red[:,:,0])    # R G B Colour
```

Out[25]:
```
<matplotlib.image.AxesImage at 0x2a9de391bd0>
```

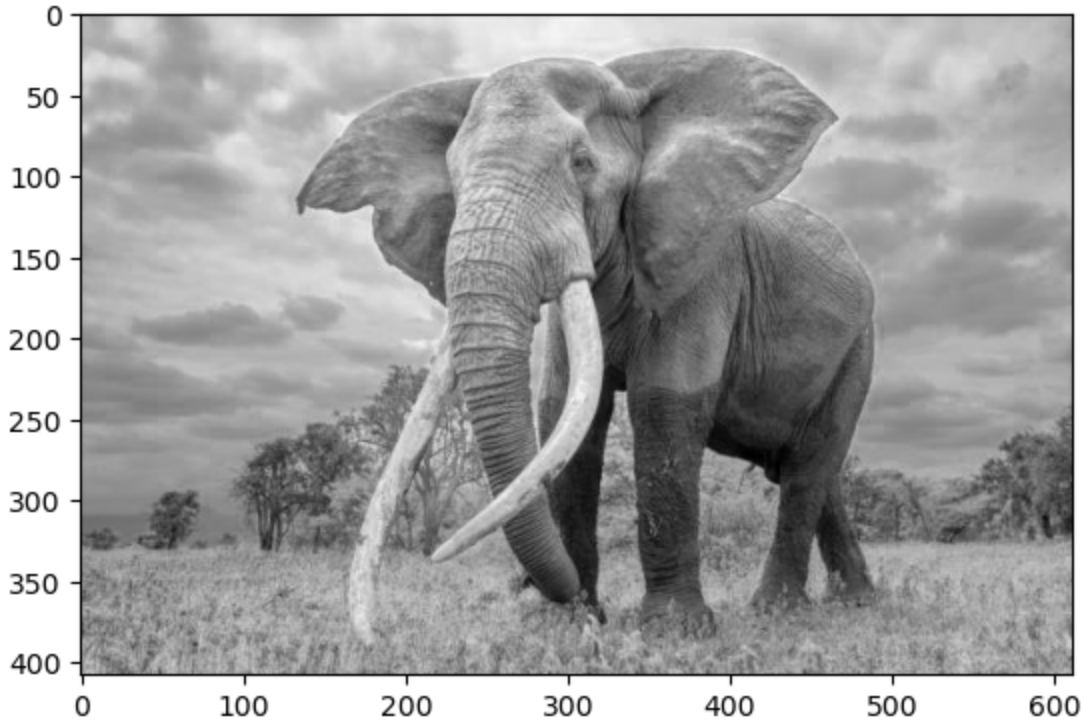


In [26]:
```
elephant_red[:,:,0]
```

Out[26]:
```
array([[195, 191, 186, ..., 182, 181, 181],
       [193, 189, 185, ..., 181, 181, 180],
       [191, 188, 183, ..., 180, 180, 179],
       ...,
       [149, 141, 158, ..., 167, 160, 151],
       [164, 172, 162, ..., 188, 183, 182],
       [147, 179, 159, ..., 156, 162, 176]], shape=(408, 612), dtype=uint8)
```

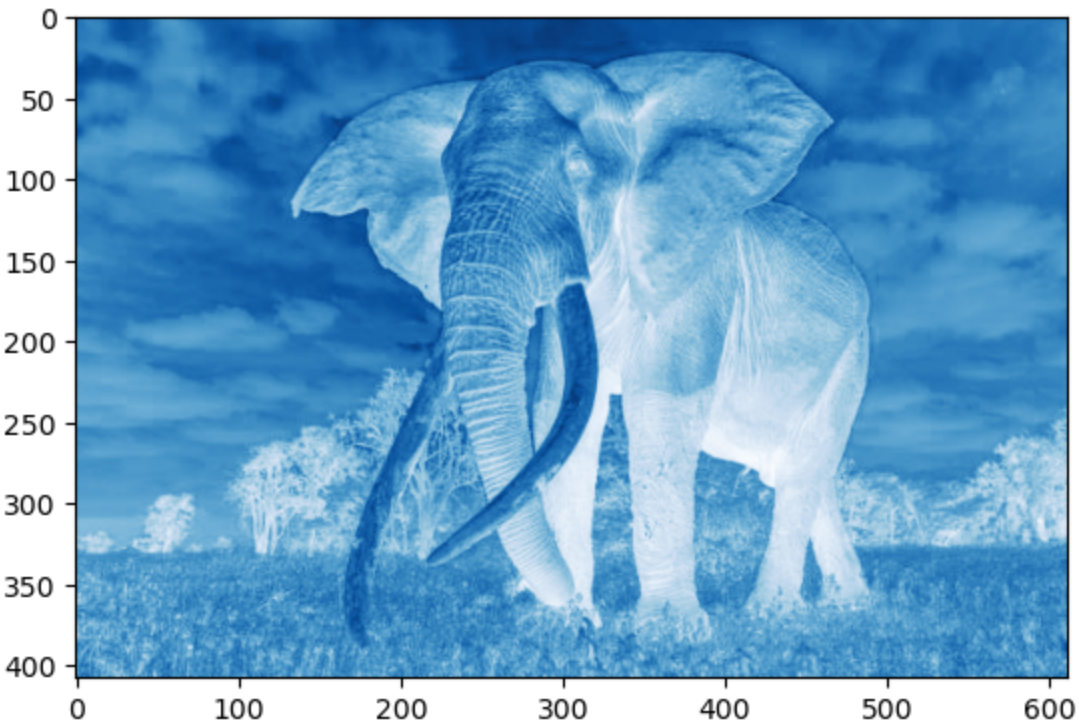
In [27]:
```
plt.imshow(elephant_red[:,:,0], cmap= 'gray')
```

Out[27]:
```
<matplotlib.image.AxesImage at 0x2a9de415810>
```



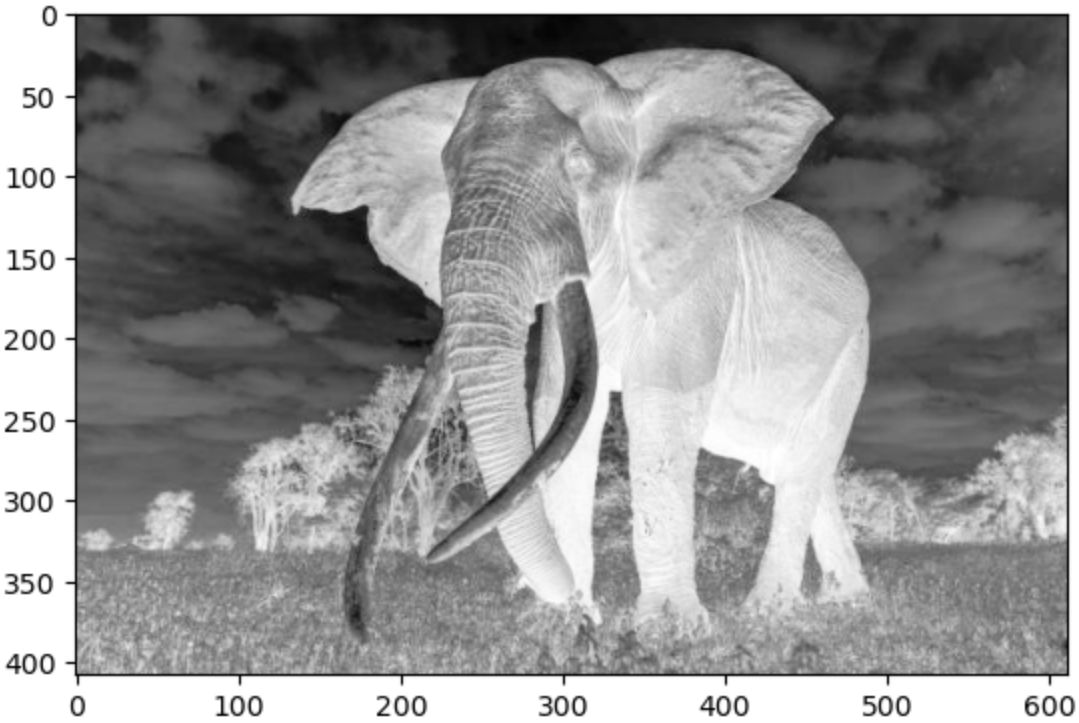
In [28]:
```
plt.imshow(elephant_red[:,:,0], cmap= 'Blues')
```

Out[28]:
```
<matplotlib.image.AxesImage at 0x2a9de4a5590>
```



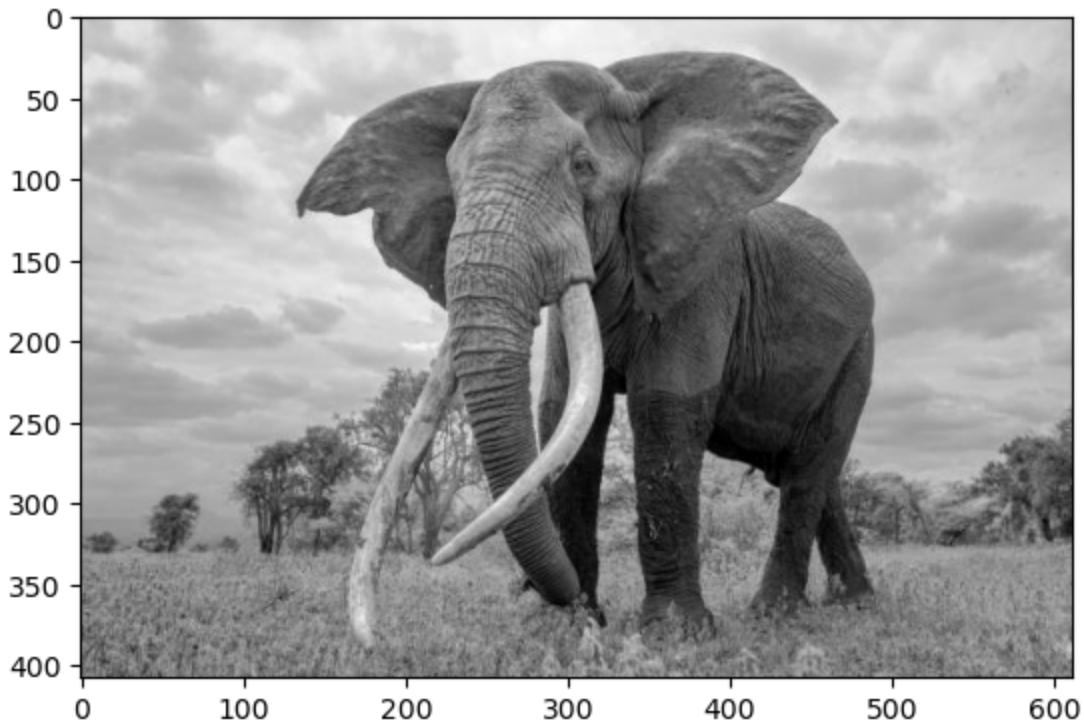
In [29]:
```
plt.imshow(elephant_red[:,:,1], cmap= 'Greys')
```

Out[29]: <matplotlib.image.AxesImage at 0x2a9de5251d0>



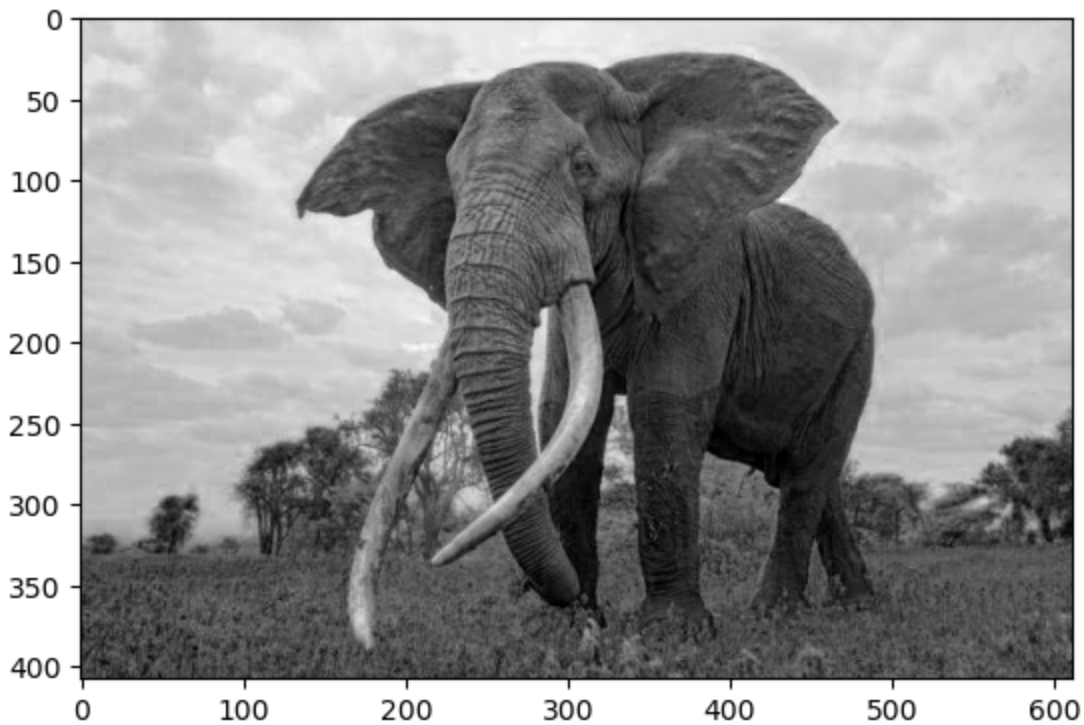
In [30]:
```
plt.imshow(elephant_red[:,:,1], cmap= 'grey')
```

Out[30]: <matplotlib.image.AxesImage at 0x2a9df578e10>



In [31]:
```
plt.imshow(elephant_red[:,:,2], cmap= 'grey')
```

Out[31]:
```
<matplotlib.image.AxesImage at 0x2a9df5fca50>
```



In [32]:
```
elephant_red[:,:,0]
```

Out[32]:
```
array([[195, 191, 186, ..., 182, 181, 181],
       [193, 189, 185, ..., 181, 181, 180],
       [191, 188, 183, ..., 180, 180, 179],
       ...,
       [149, 141, 158, ..., 167, 160, 151],
       [164, 172, 162, ..., 188, 183, 182],
       [147, 179, 159, ..., 156, 162, 176]], shape=(408, 612), dtype=uint8)
```

In [33]:
```
elephant_red[:,:,1]
```

Out[33]:
```
array([[207, 203, 198, ..., 196, 195, 195],
       [205, 201, 197, ..., 195, 195, 194],
       [203, 200, 195, ..., 194, 194, 193],
       ...,
       [ 99,  92, 106, ..., 124, 119, 111],
       [115, 123, 111, ..., 140, 137, 139],
       [ 98, 130, 108, ..., 104, 112, 129]], shape=(408, 612), dtype=uint8)
```

In [34]:
```
elephant_red[:,:,2]
```

Out[34]:
```
array([[223, 219, 214, ..., 207, 206, 206],
       [221, 217, 213, ..., 206, 206, 205],
       [219, 216, 211, ..., 205, 205, 204],
       ...,
       [ 62,  52,  66, ...,  73,  67,  59],
       [ 75,  82,  68, ...,  91,  87,  88],
       [ 57,  87,  63, ...,  54,  61,  77]], shape=(408, 612), dtype=uint8)
```

In [35]:
```
elephant_red[:,:,1] = 0
```

In [36]:
```
elephant_red[:,:,1]
```

Out[36]:
```
array([[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       ...,
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0]], shape=(408, 612), dtype=uint8)
```

In [37]:
```
plt.imshow(elephant_red)
```

Out[37]:
```
<matplotlib.image.AxesImage at 0x2a9df684a50>
```

In [38]:
```
elephant_red[:,:,2]
```

Out[38]:
```
array([[223, 219, 214, ..., 207, 206, 206],
       [221, 217, 213, ..., 206, 206, 205],
       [219, 216, 211, ..., 205, 205, 204],
       ...,
       [ 62,  52,  66, ...,  73,  67,  59],
       [ 75,  82,  68, ...,  91,  87,  88],
       [ 57,  87,  63, ...,  54,  61,  77]], shape=(408, 612), dtype=uint8)
```

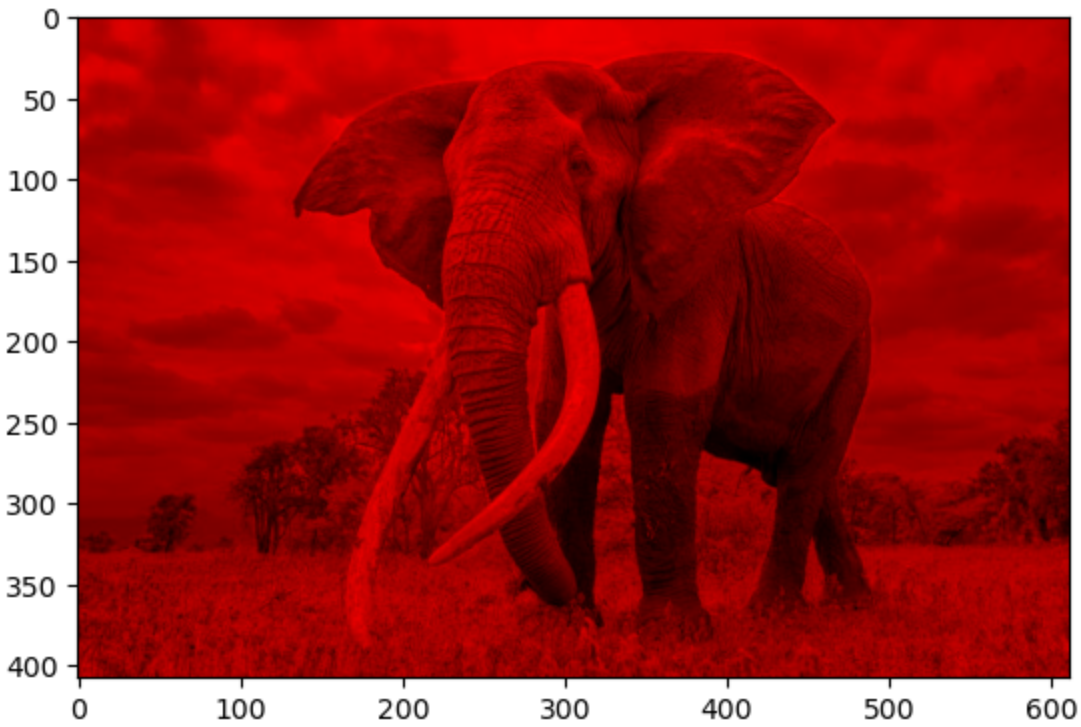
In [39]:
```
elephant_red[:,:,2] = 0
```

In [40]:
```
elephant_red[:,:,2]
```

Out[40]:
```
array([[0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       ...,
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0],
       [0, 0, 0, ..., 0, 0, 0]], shape=(408, 612), dtype=uint8)
```

In [41]:
```
plt.imshow(elephant_red)
```

Out[41]:
```
<matplotlib.image.AxesImage at 0x2a9df7082d0>
```



In [42]:
```
elephant_arr
```

Out[42]:
```
array([[[195, 207, 223],
        [191, 203, 219],
        [186, 198, 214],
        ...,
        [182, 196, 207],
        [181, 195, 206],
        [181, 195, 206]],

       [[193, 205, 221],
        [189, 201, 217],
        [185, 197, 213],
        ...,
        [181, 195, 206],
        [181, 195, 206],
        [180, 194, 205]],

       [[191, 203, 219],
        [188, 200, 216],
        [183, 195, 211],
        ...,
        [180, 194, 205],
        [180, 194, 205],
        [179, 193, 204]],

       ...,

       [[149,  99,  62],
        [141,  92,  52],
        [158, 106,  66],
        ...,
        [167, 124,  73],
        [160, 119,  67],
        [151, 111,  59]],

       [[164, 115,  75],
        [172, 123,  82],
        [162, 111,  68],
        ...,
        [188, 140,  91],
        [183, 137,  87],
        [182, 139,  88]],

       [[147,  98,  57],
        [179, 130,  87],
        [159, 108,  63],
        ...,
        [156, 104,  54],
        [162, 112,  61],
        [176, 129,  77]]], shape=(408, 612, 3), dtype=uint8)
```

In [43]:
```
elephant_red
```

Out[43]:
```
array([[[195,   0,   0],
        [191,   0,   0],
        [186,   0,   0],
        ...,
        [182,   0,   0],
        [181,   0,   0],
        [181,   0,   0]],

       [[193,   0,   0],
        [189,   0,   0],
        [185,   0,   0],
        ...,
        [181,   0,   0],
        [181,   0,   0],
        [180,   0,   0]],

       [[191,   0,   0],
        [188,   0,   0],
        [183,   0,   0],
        ...,
        [180,   0,   0],
        [180,   0,   0],
        [179,   0,   0]],

       ...,

       [[149,   0,   0],
        [141,   0,   0],
        [158,   0,   0],
        ...,
        [167,   0,   0],
        [160,   0,   0],
        [151,   0,   0]],

       [[164,   0,   0],
        [172,   0,   0],
        [162,   0,   0],
        ...,
        [188,   0,   0],
        [183,   0,   0],
        [182,   0,   0]],

       [[147,   0,   0],
        [179,   0,   0],
        [159,   0,   0],
        ...,
        [156,   0,   0],
        [162,   0,   0],
        [176,   0,   0]]], shape=(408, 612, 3), dtype=uint8)
```

In [44]:
```
elephant_img
```

Out[44]:

In [45]:
```
arr1 = np.asarray(elephant_img)
```

In [46]:
```
arr1
```

Out[46]:
```
array([[[195, 207, 223],
        [191, 203, 219],
        [186, 198, 214],
        ...,
        [182, 196, 207],
        [181, 195, 206],
        [181, 195, 206]],

       [[193, 205, 221],
        [189, 201, 217],
        [185, 197, 213],
        ...,
        [181, 195, 206],
        [181, 195, 206],
        [180, 194, 205]],

       [[191, 203, 219],
        [188, 200, 216],
        [183, 195, 211],
        ...,
        [180, 194, 205],
        [180, 194, 205],
        [179, 193, 204]],

       ...,

       [[149,  99,  62],
        [141,  92,  52],
        [158, 106,  66],
        ...,
        [167, 124,  73],
        [160, 119,  67],
        [151, 111,  59]],

       [[164, 115,  75],
        [172, 123,  82],
        [162, 111,  68],
        ...,
        [188, 140,  91],
        [183, 137,  87],
        [182, 139,  88]],

       [[147,  98,  57],
        [179, 130,  87],
        [159, 108,  63],
        ...,
        [156, 104,  54],
        [162, 112,  61],
        [176, 129,  77]]], shape=(408, 612, 3), dtype=uint8)
```

In [47]:
```
type(arr1)
```

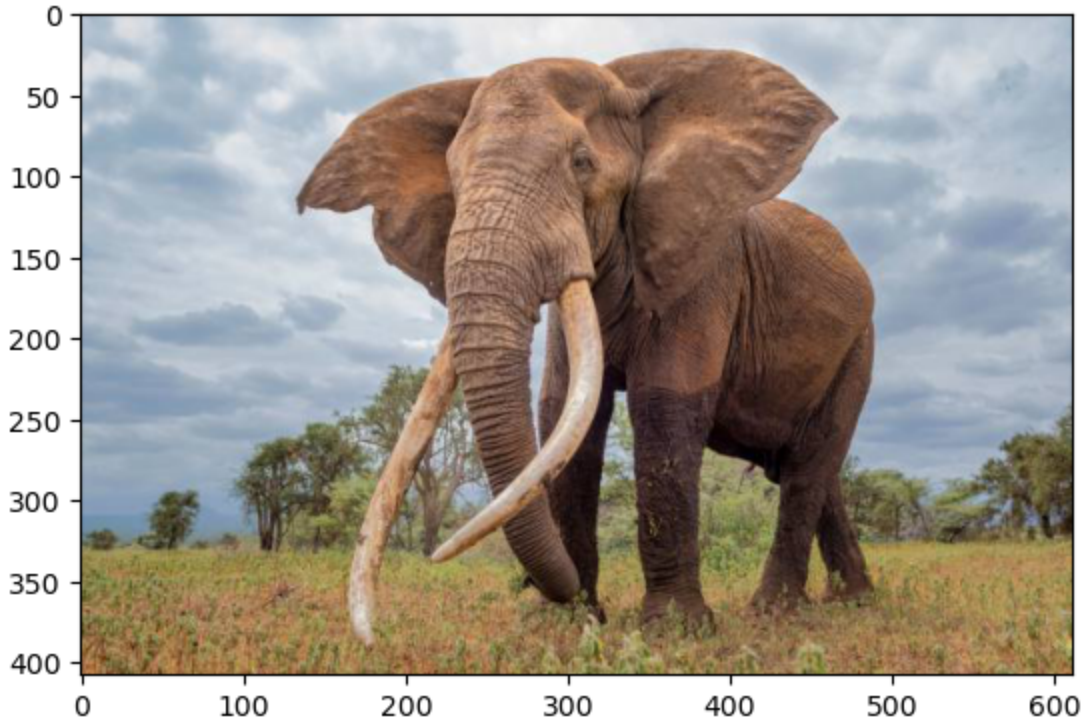
Out[47]:
```
numpy.ndarray
```

In [48]:
```
arr1.shape
```

Out[48]: (408, 612, 3)

In [49]:
```
plt.imshow(arr1)
```

Out[49]: <matplotlib.image.AxesImage at 0x2a9df76bed0>

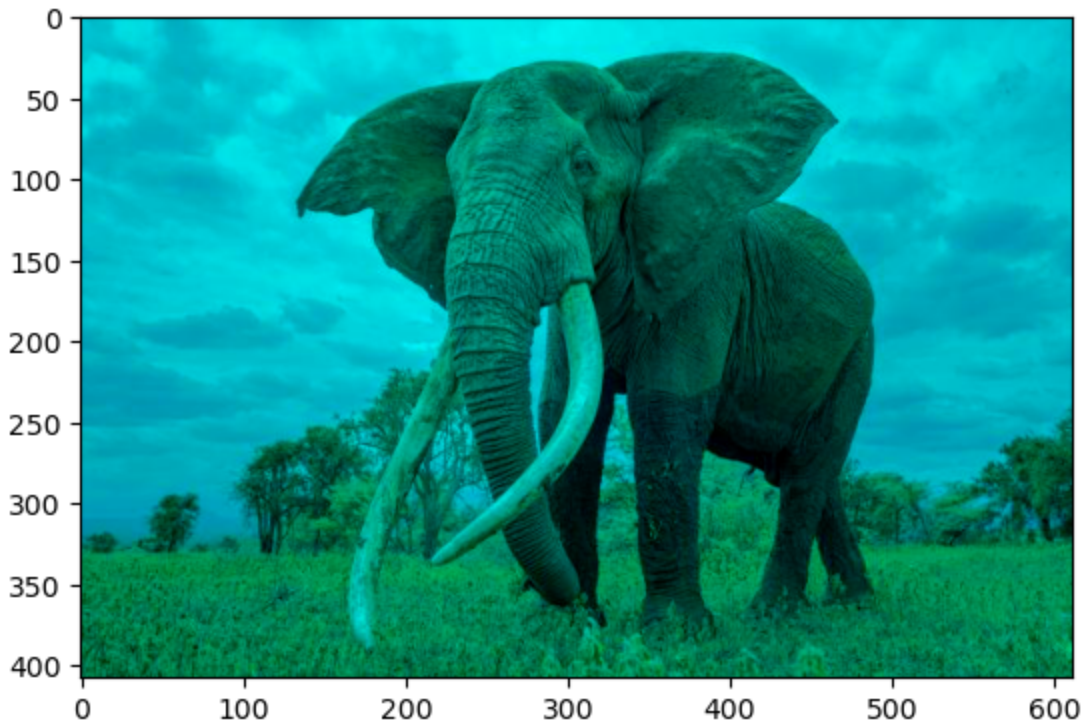


In [50]:
```
elephant_img1 = arr1.copy()
```

In [51]:
```
elephant_img1[:,:,0] = 0
```

In [52]:
```
plt.imshow(elephant_img1)
```

Out[52]: <matplotlib.image.AxesImage at 0x2a9df7ebb10>



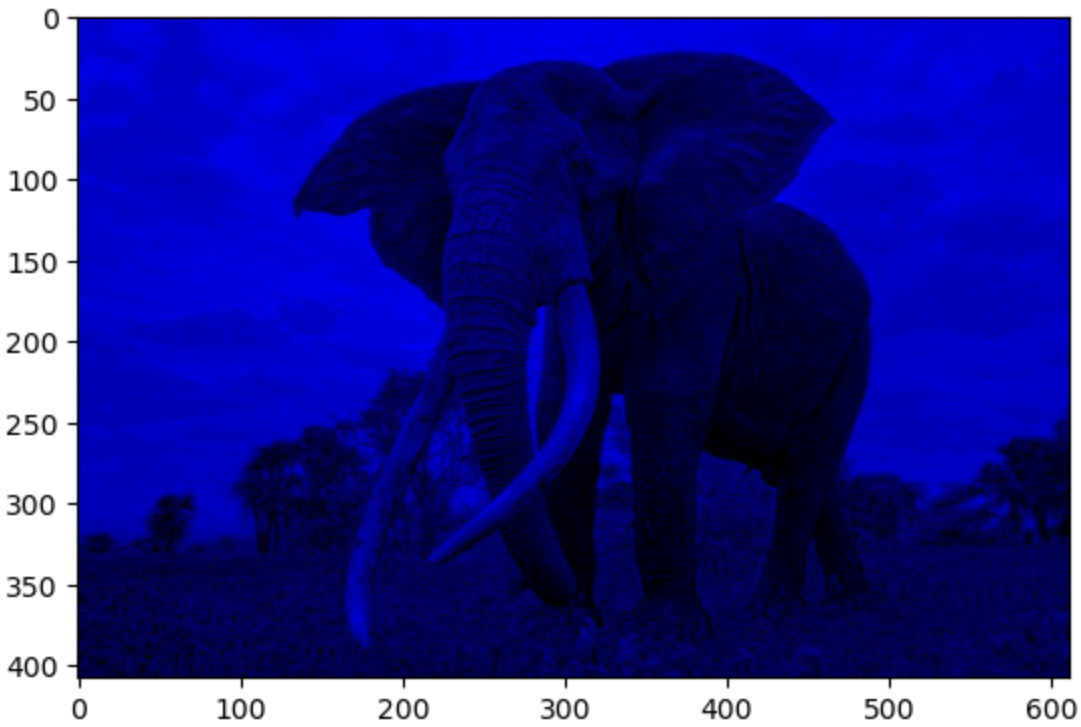
In [53]:
```
elephant_img1[:,:,1]
```

Out[53]:
```
array([[207, 203, 198, ..., 196, 195, 195],
       [205, 201, 197, ..., 195, 195, 194],
       [203, 200, 195, ..., 194, 194, 193],
       ...,
       [ 99,  92, 106, ..., 124, 119, 111],
       [115, 123, 111, ..., 140, 137, 139],
       [ 98, 130, 108, ..., 104, 112, 129]], shape=(408, 612), dtype=uint8)
```

In [54]:
```
elephant_img1[:,:,1] = 0
```

In [55]:
```
plt.imshow(elephant_img1)
```

Out[55]:
```
<matplotlib.image.AxesImage at 0x2a9df873750>
```



In [ ]: